An IoT Application for Detection and Monitoring of Manhole

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ABSTRACT: In today's world manhole problems in the populated areas is the major issue. The probability of rate of accidents on roads is increasing gradually due to improper and damaged manholes, this leads to heavy risk for the individual safety. The good maintenance of manhole is a sign of good and cleaner cities. There are modern methods to solve this problem, for this we are using Internet of Things (IoT) technology. The IoT consists of real-life objects, communication devices attached to sensor networks in order to provide communication and automated actions between real world and information world. Manhole Detection and monitoring is mainly based on detecting the manholes which are opened due to breakage of lid, overflow of sewage water or rainwater during heavy rainfall, sudden changes in temperature level and gas levels. To prevent this, different sensors are placed in the manhole to sense different parameters which are sent to the managing authorities via smart phone using GSM module. Based on the received information the municipal administration will take the necessary action to resolve the manhole problem. In this paper we have explained and demonstrated this problem using IoT.

1.INTRODUCTION

Manholes play an essential part in maintaining city sewage systems. They serve not only as maintenance access points, but also act as anchors in sewer system design. Sewage systems have been around almost since the dawn of civilization. A manhole is an opening to a confined space such as a shaft, utility vault, or large vessel. Manholes are also a key component in sewage system design. These service pipes are used to join different parts of sewers together. Manholes are often used as an access point for an underground public utility, allowing inspection, maintenance, and system upgrades. Most underground services have manholes, including water, sewers, electricity, storm drains and gas. In this paper, we are discussing about the manholes which are used for sewage and drainage systems. Most of the cities adopted the underground drainage system and it is the duty of managing station (Municipal Corporation) to maintain cleanliness of the cities.

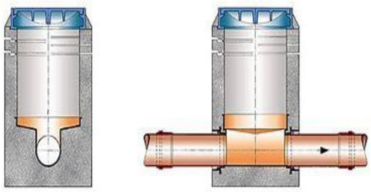

Fig 1.1 Shallow Manhole

According to the traffic of the sewage or depending on the area there are different types of manhole systems, they are[14]-[16]:

- Shallow Manhole
- Normal Manhole
- Deep Manhole

Shallow Manhole: If the depth of the manhole is 2 to 3 feet down, it is known as a shallow manhole. These are found at the beginning of a sewer branch as shown in Fig.1.1.
Normal Manhole: In this type of manhole, the depth ranges from 4 to 5 feet. Most manholes are round, but they may be rectangular or square instead as shown in Fig.1.2


Fig 1.2 Normal Manholes

Deep Manhole: In this type of manhole, the depth greater than 5 feet. They have heavy covers and have a gradually increasing diameter as shown in Fig.1.3


Fig.1.3 Deep Manholes

At present main problem in society is poor drainage system and many accidents occur due to the lack of communication about the breakage of the manhole If the drainage maintenance is not proper the pure water gets contaminated with drainage water and infectious diseases may get spread. The drainage gets blocked during rainy season, it will create problem for routine life such as traffic may get jammed, the environment becomes dirty, and totally it upsets the public. The poor drainage systems are main cause for spread of diseases like typhoid, dengue, hepatitis. The main reason for diseases is staging of sewage water on the roads. Due to this, diseases may spread by direct or indirect contact. This happens because of the ignorance or not able to know about the problem by the Municipal authorities. Suppose if there should be a facility which would be there in Municipal Corporation (managing station) that the officials come to know immediately after blocking of drainage in the manhole or breakage of the lid, they may respond according to the situation. Hence it can reduce the numerous accidents. So we require a device that monitors the manhole and detects whenever there is a change i.e., blockage of sewage water due to heavy rains, when the lid is opened or broke by other factors, increase or change in the temperature or gas.

In the modern technology we can be able to monitor and detect all these factors by using different technologies. One of them is by using IoT technology for monitoring all these factors. The IoT describes the network of physical objects(things) that are embedded with sensors, software and other technologies for the purpose of connecting and exchanging data with other devices and systems over the internet. These devices range from household to industrial tools. Manhole detection and monitoring system which is based on IoT is able to detect whether the lid is opened or water level, gas and temperature in the manhole increasing and sends the message to the managing authorities, so that certain measures will be taken by them. There are different sensors like tilt sensor which is used to detect the opening or breakage of the lid, water level sensor which sends the information of the water level in the manhole, gas sensor that predicts the gas and temperature sensor that predicts the change in the level of the temperature, and it sends the data to the managing authorities. With this device a lot of accidents can be prevented, and it helps in maintaining the city clean.

2. LITERATURE SURVEY:

From [1], Today the entire world contending to construct brilliant urban areas. Savvy underground foundation could be a significant part in execution of shrewd urban areas. Seepage framework observing assumes significant part keep the town spotless and solid. Since, labor isn't adequate for constant observing and to differentiate minute issues. So, we are hoping on the IOT frameworks. The proposed framework is minimal expense, low upkeep, IoT based constant which alarms the overseeing station through the messages when any issues in sewer vent is distinguished. In [1] , they proposed a minimal expense and special sewer vent interruption identification framework with notice stages which shields basic foundation. The proposed framework uses an Arduino Uno and different sensors to trigger the interruption at right time before the copper links are taken.

From [2], underground drainage systems are currently most widespread in large places such as Mumbai. Because of the increasing population there will be a lot of waste. If the drainage is not properly maintained, pure water might get contaminated by drainage water, resulting in the spread of infectious diseases. Since decades, a lot of accidents have occurred due to lack of proper gas leakage systems, using gas sensors, the system is designed to detect gases such carbon monoxide, hydrogen sulphide and methane.

From [3], monitoring of sewage outlet system is very important in recent times. The presence of clog in drainage system results to overflow in streets which results in spread of diseases. In recent survey, it is found that hundreds of people died every year due to manual cleaning of drainage system. The main intention of this project is to monitor and remove the clog in the drainage system before the water overflow. This project utilized a water level sensor to identify the clog in the manhole.

From [4], most of the Indian cities have adopted an underground waste management system, the complexity of this system is more hence it is critical that this system must function properly. If the sewage framework isn't properly maintained, we could face serious problems. As a result, numerous technologies have emerged to identify, track, and manage these sewage networks This challenge focuses on the ability to plan for underground observation using multiple methods.

From [5], drainage is the system or process by which water, sewage or other liquids are drained from one place to another and to maintain the proper function of drainage, its condition should be monitored regularly. But it is very difficult to monitor all area manually where a human cannot reach. To solve all these problems caused, in [5] the authors have developed and implemented the system using wireless sensor network.

From [6], in India the waste framework is one in every of the numerous issues, thanks to the helpless support of the sewage framework the sewage water is flooded and obstruct within the city and now so blends within the drinkable which harms the medical issue of the surrounding individuals, to defeat this issue we are proposing the model called Drainage Overflow Monitoring System (DOMS). This proposed framework will screen the water level and gas level within the sewage framework.

From [7], in urban areas, the Internet of Things (IoT) could be a significant viewpoint which gives simple and provides solution for the public problems. Today urban India is facing trouble with drain or sewage water, most of waste enters lakes, waterways and groundwater without treating, this is due to the ignorance of the municipal authorities in cities. This paper gives an IoT based framework to deal with this issue.

From [8], Certain measures have recently been taken to improve the health conditions and hygiene of the country. Many drainages or sewage problems lead to poor sanitation, leading to human and life-threatening diseases. You can overcome these issues by innovative and effective ways. The humidity and dryness sensors can be used to detect overflow. To detect dangerous gases such as methane, gas sensors can be detected, and temperature can be detected by temperature sensors. It also saves the lives of medical staff. Safety for sanitary workers also involved in it.

From [9], This paper presents the use of savvy sensors for observing both the stormwater and channel or waste product organizations of the Scientific grounds of the University of Lille within the North of France. This ground represents a town of around 25 000 clients. for every organization, the paper presents the checking framework, examination of the recorded information and the way this investigation caused upgrading our comprehension of the organization working even as its improvement.

From [10], In India, the introduction of drainage systems is very important. There are many places that are not clean and are clogged Drainage due to mishandling of sewer cleaners. clever Sabotage Drainage System - Essentials multiple infrastructures. If the drainage system cannot be maintained clean water can be contaminated with wastewater. It spreads the epidemic. Using an ultrasonic sensor, the information will be sent to the ESP32 microcontroller whenever a block is found.

From [11], Manhole cover (MC) failure is on the upward push and typically impacts the protection and economic system of the society. That is why the need for fully automatic tracking system has come to be very critical. Automatic tracking of MC is part of the development of smart towns (SC) and internet of things (IoT) which can be the objectives for cutting-edge governments to govern and screen the resources in cities. This paper is a survey study on the MC troubles, imparting a complete category with analysis for those troubles based on the environmental effect and which include the modern-day tracking techniques.

From [12], now a days manhole problems within the populated towns is the one of big troubles. Establishing of manholes due to breakage of manhole cover, manhole explosions are essential threat in recent days. Manhole detection and alerting is especially based totally on detecting the manholes that are opened due to overflow of sewage / rain water. To keep away from such incidents even before it may affect the public, an alerting device is built in which the buzzer signals and sends the sensed information to the handling government using GSM strategies. So, they could take precautions to shut the manhole for public safety.

3.SYSTEM MODEL:

3.1 BLOCK DIAGRAM

THE SYSTEM CONSIST OF BOTH HARDWARE AND SOFTWARE COMPONENTS:

- HARDWARE COMPONENTS:
  - Arduino UNO
  - Temperature sensor
  - Water flow sensor
  - Tilt sensor
  - Gas sensor
  - LCD
  - GSM module
  - Buzzer
  - Power supply
- SOFTWARE COMPONENTS:
  - Arduino

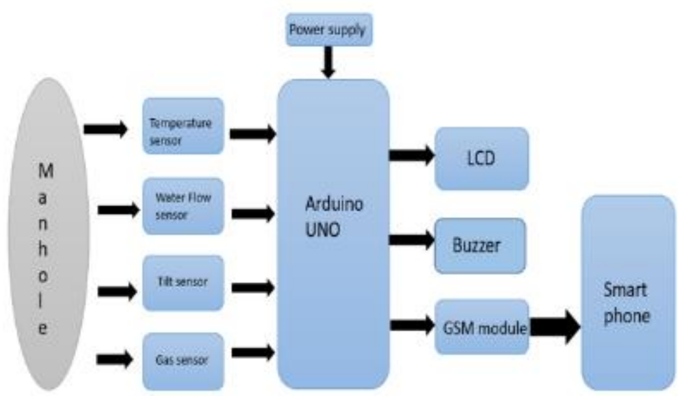

Fig.2 Block diagram for manhole detection and monitoring system

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3.2 HARDWARE COMPONENTS:

ARDUINO UNO: Arduino is a open -source prototyping platform in electronics based easy to-use hardware and software. It is a low cost and flexible. This board contains USB interface i.e., USB cable.

TEMPERATURE SENSOR: Temperature sensors plays a critical role in detecting a specific temperature in surroundings.

WATER LEVEL SENSOR: Water flow sensor consists of copper body, and it detects the water level by change occurred in the resistance of the device due to change in the distance from top of the sensor to the surface of the water.

GAS SENSOR: Gas sensor is also known as gas detector. They are electronic devices that detect and identify different types of gases.

TILT SENSOR: The tilt sensor is a device used for measuring the tilt of an object in multiple axes with reference to an absolute level plane.

LCD: LCD Stands for liquid crystal display

GSM MODULE: A **GSM** (Global System for Mobile Communications, originally Groupe Special Mobile) is a hardware device that uses GSM mobile telephone technology to provide a data link to a remote network.

BUZZER: Buzzer is an audio signaling device.

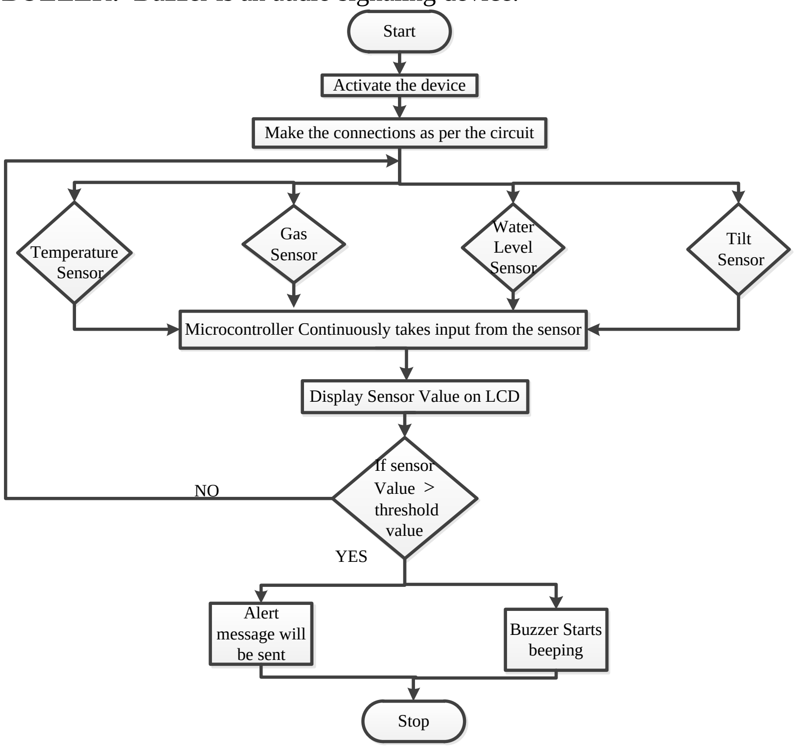

Fig.3.Flowchart for the operation of system model

3.3 SOFTWARE DESCRIPTION:

Arduino microcontrollers are single-board computers that are popular in both the hobby and professional markets because to their ease of use and high performance. The hardware is affordable, and software development is free, because Arduino is an open-source initiative. It features 2 KB of RAM, 32 KB of flash memory for program storage, and 1 KB of EEPROM for parameter storage. At a clock speed of 16MHz, it runs around 300,000 lines of C source code per second. The assessment board has 14 digital I/O pins and 6 analogue input pins. When the application is run without being connected to the host computer, a USB connector is provided as well as a DC power jack for attaching an external 6-20V power source to the I/O pins. The accompanying header connector is a 22g single wire connector.

5. OPERATION OF THE SYSTEM

Fig.4: Manhole Detection using Water Level Sensor

An “IoT based manhole detection and monitoring system” will not only help in maintaining the proper health and safety of the city but also in reducing the work of government personnel. For this system, the connections were made as per the block diagram shown in the Fig.2. Various types of sensors like water level sensor, tilt sensor, temperature sensor and gas sensor are interfaced with microcontroller Arduino Uno in order to make the system smart. Then the output ports of Arduino Uno are connected with other components like LCD, GSM module, buzzer. Firstly, the power supply is provided to the Arduino Uno board in order to start the working of the system or a device. A 5v stabilized DC power supply is provided as an input to the Arduino board because this microcontroller is a low power device and can be damaged if the input exceeds more than 5v. The detailed flowchart for the operation of the system model is shown in Fig.3.

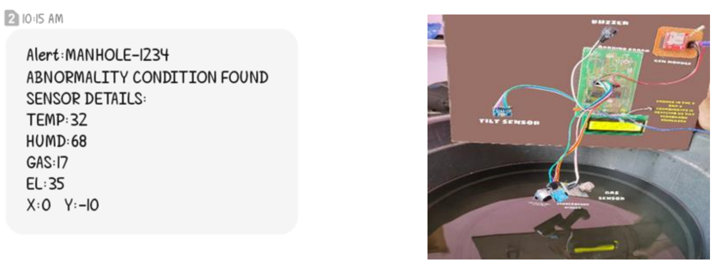

Fig.5: Manhole Detection using Tilt Sensor

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After giving the power supply, the device starts working. Firstly, the device senses all the sensors simultaneously, if there is an increase in temperature then it can be detected by using temperature sensor. The temperature sensor detects the temperature and sends the digital data values as an input to the microcontroller. This microcontroller checks the data continuously and if the data value is more than the threshold voltage then microcontroller sends the alert message to the managing authorities through smart phone by using GSM module. The message will be displayed on the LCD and buzzer starts alarming. If there is a raise in water level flow then it can be detected by using water level sensor. It is shown in Fig.4. Water level value w.r.t water level sensor less than 5cm indicates that a Manhole is detected. In the received message, we obtained water level as 4cm. If the lid is tilted, then it can be detected by using tilt sensor. The tilt sensor detects the orientation of a lid and gives its output. It is shown in Fig.5. Tilt in the Manhole is detected when the coordinates are X>5 or X<-5 and Y>5 or Y<-5. In the received message, Y- Coordinate is -10. It indicates that a manhole has been detected. If the manhole is filled up with smoke or gases then it will be detected by using gas sensor. This sensor senses the gas and sends the output. It is shown in Fig.6. If the gas sensor value is above 50ppm, then manhole is detected. In the received message, we obtained gas level as 56ppm. Municipal authorities will take necessary action to resolve the problem.

By this method, the officials could easily get notified about the problem and will take appropriate steps to solve the issue within a less interval of time. Also, Arduino Uno updates the live values of all the sensors in the manholes using IoT by displaying the message on the LCD, which gives a clarity to the in-charge about the problem occurred.

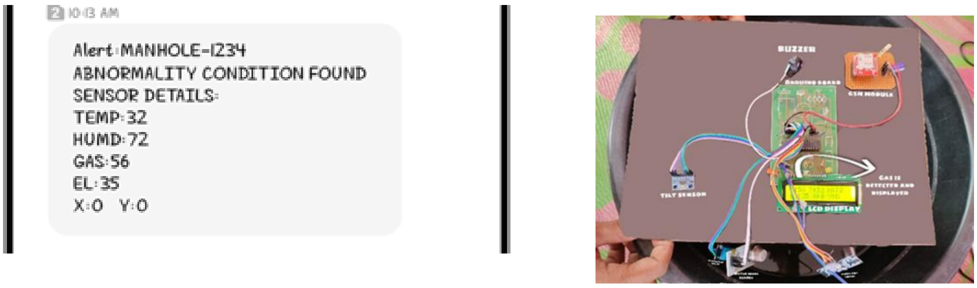

Fig.6: Manhole Detection using Gas Sensor

6. CONCLUSION:

In these days manhole monitoring is challenging problem. This paper provides different methods for monitoring and detecting underground drainage system. It explains various applications like underground drainage and manhole identification in real time. Various parameters like temperature, toxic gases, flow and level of water and lid breakage are being monitored and updated to the managing authorities using the Internet of Things. This enables the person in-charge to take the necessary actions regarding the same. In this way the unnecessary trips on the manholes are saved and can only be conducted when required, this reduces a lot of risks caused to the mankind. Also, real time update to the managing authorities helps in maintaining the regularity in drainage check thus avoid the hazards.

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